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4.9 Chemistry of the Atmosphere

AQA GCSE Chemistry (8462) — Paper 2

The Earth's atmosphere is dynamic and forever changing. The causes of these changes are sometimes man-made and sometimes part of many natural cycles. Scientists use very complex software to predict weather and climate change, as there are many variables that can influence this. The problems caused by increased levels of air pollutants require scientists and engineers to develop solutions that help to reduce the impact of human activity. This whole topic is common to both Chemistry and Combined Science; the evaluation skills (interpreting evidence, uncertainty, peer review) are tested heavily in extended-answer questions.

4.9.1 The Composition and Evolution of the Earth's Atmosphere

4.9.1.1 The Proportions of Different Gases in the Atmosphere

For **200 million years**, the proportions of different gases in the atmosphere have been much the same as they are today.

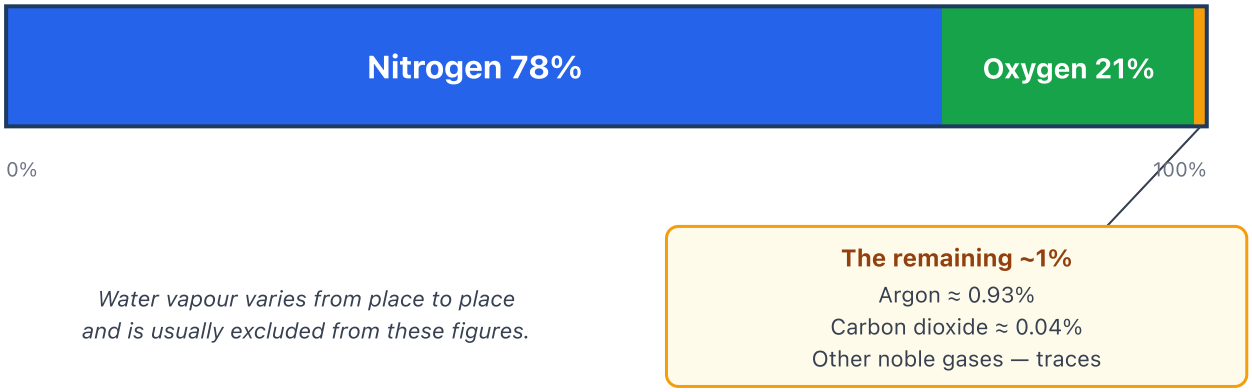
> The Earth's atmosphere today is approximately:
> • about **four-fifths (approximately 80%) nitrogen**;
> • about **one-fifth (approximately 20%) oxygen**;
> • **small proportions of various other gases**, including carbon dioxide, water vapour and noble gases.

Gas	Formula	Approximate % by volume (dry air)
Nitrogen	N_2	78%
Oxygen	O_2	21%

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Argon (a noble gas)	Ar	0.93%
Carbon dioxide	CO_2	0.04% (about 420 parts per million)
Other noble gases (neon, helium, krypton, xenon)	Ne, He, Kr, Xe	Traces
Water vapour	H_2O	Variable (0–4%) — not included in "dry air"

Composition of dry air today



Exam Tip: AQA's mark schemes use "about four-fifths nitrogen and about one-fifth oxygen". If a question asks for percentages, 78% and 21% (or "about 80%" and "about 20%") are both accepted. A frequent mistake is to say the atmosphere is "mostly oxygen" — it is mostly **nitrogen**.

Worked Example 1 — Using Proportions

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Question: A room contains 50 m³ of air. Estimate the volume of
          oxygen in the room.

Oxygen ≈ one-fifth (20%) of air
Volume of oxygen = 50 × 0.20 = 10 m³
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```
(Using 21%: 50 × 0.21 = 10.5 m³ – either answer is acceptable
 if the working is shown.)
```

4.9.1.2 The Earth's Early Atmosphere

Theories about what was in the Earth's early atmosphere and how the atmosphere was formed have changed and developed over time. **Evidence for the early atmosphere is limited** because of the time scale of **4.6 billion years** — there are no direct samples, so scientists rely on evidence such as the composition of gases from volcanoes today, ancient rocks, and the atmospheres of other planets.

One theory suggests:

- During the **first billion years** of the Earth's existence there was **intense volcanic activity** that released gases forming the early atmosphere, and water vapour that **condensed to form the oceans**.
- At the start of this period the Earth's atmosphere may have been like the atmospheres of **Mars and Venus** today, consisting of mainly **carbon dioxide** with **little or no oxygen gas**.
- Volcanoes also produced **nitrogen**, which gradually built up in the atmosphere, and there may have been small proportions of **methane** and **ammonia**.
- When the oceans formed, **carbon dioxide dissolved in the water** and **carbonates were precipitated**, producing sediments, which **reduced** the amount of carbon dioxide in the atmosphere.

Gas	Early atmosphere	Today	Reason for change
Carbon dioxide	Very high (main gas)	~0.04%	Dissolved in oceans; locked into carbonate rocks and fossil fuels; used in photosynthesis.
Oxygen	Little or none	~21%	Produced by photosynthesis in algae and plants.
Nitrogen	Small amount	~78%	Released by volcanoes; very unreactive so it gradually built up.

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Water vapour	High	Variable, small	Condensed as the Earth cooled to form the oceans.
Methane, ammonia	Small amounts (possibly)	Traces	Reacted with oxygen once it appeared.

> **Exam Tip:** Questions often give you information about a **different, unfamiliar** theory of the early atmosphere (e.g. one involving comets) and ask you to interpret it. You are not expected to know these alternatives — just use the information given, and explain **why the evidence is limited**: it was 4.6 billion years ago and nobody was there to measure it.

4.9.1.3 How Oxygen Increased

Algae and plants produced the oxygen that is now in the atmosphere by **photosynthesis**:

> `carbon dioxide + water → glucose + oxygen`
> $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
>
> (Energy is absorbed from light; the reaction is **endothermic**.)

- **Algae** first produced oxygen about **2.7 billion years ago**.
- Soon after this, oxygen appeared in the atmosphere.
- Over the **next billion years**, **plants evolved** and the percentage of oxygen gradually increased to a level that enabled **animals to evolve**.

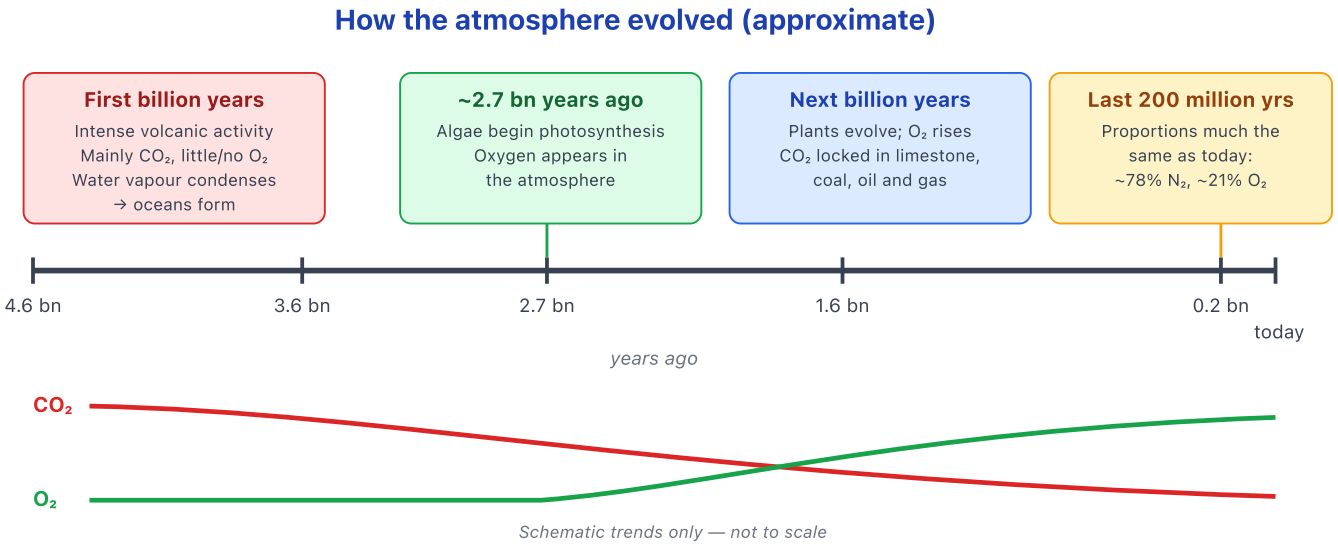
4.9.1.4 How Carbon Dioxide Decreased

Carbon dioxide was removed from the early atmosphere in several ways:

1. **Dissolving in the oceans** — then precipitated as **carbonates** that formed sediments.
2. **Photosynthesis** by **algae and plants** — CO_2 was converted into glucose and other carbon compounds in living things.

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3. **Formation of sedimentary rocks** — the shells and skeletons of marine organisms (made of **calcium carbonate**) built up on the sea bed and formed **limestone**.
4. **Formation of fossil fuels** that contain carbon:
   - **Coal** — a sedimentary rock made from thick plant deposits buried and compressed over millions of years.
   - **Crude oil and natural gas** — formed from deposits of **plankton** buried in **mud** on the sea bed, subjected to heat and pressure.



Exam Tip: For a 4- or 6-mark question on "how the atmosphere changed", structure your answer around the **three gases**: CO_2 went **down** (dissolved in oceans → carbonates/sediments; photosynthesis; limestone; fossil fuels), O_2 went **up** (photosynthesis by algae then plants), N_2 went **up** (volcanoes; unreactive so it accumulated). Give a balanced photosynthesis equation for an extra mark.

4.9.2 Carbon Dioxide and Methane as Greenhouse Gases

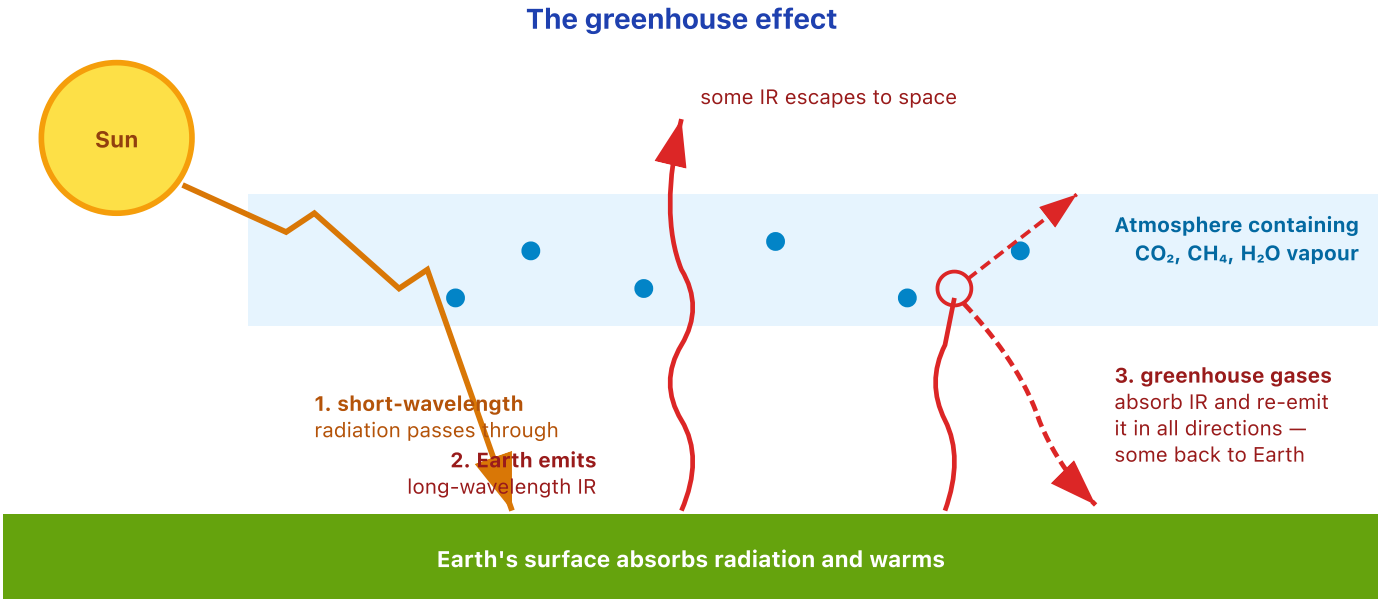
4.9.2.1 Greenhouse Gases

Greenhouse gases in the atmosphere **maintain temperatures on Earth high enough to support life**. Without them, the Earth's average surface temperature would be about –18 °C instead of about +15 °C.

Water vapour, **carbon dioxide** and **methane** are greenhouse gases.

How the Greenhouse Effect Works

1. Electromagnetic radiation from the Sun, at mostly **short wavelengths** (visible light and ultraviolet), **passes through** the atmosphere.
2. The Earth's surface **absorbs** this radiation and warms up.
3. The warm Earth **emits** (radiates) energy as **longer-wavelength infrared radiation**.
4. **Greenhouse gases absorb** some of this infrared radiation and **re-emit** it in all directions — including back towards the Earth.
5. This energy is **trapped** in the lower atmosphere, which keeps the Earth warm.



Exam Tip: The key words examiners look for are: **short wavelength** in, **long wavelength (infrared)** out, greenhouse gases **absorb** the long-wavelength radiation and **re-emit** it. Do **not** say greenhouse gases "reflect" heat back to Earth, and do not confuse the greenhouse effect with the hole in the ozone layer — they are completely different problems.

4.9.2.2 Human Activities Which Contribute to an Increase in Greenhouse Gases

Some human activities increase the amounts of greenhouse gases in the atmosphere. This is called the **enhanced greenhouse effect**.

Greenhouse gas	Human activities that increase it
Carbon dioxide	• **Burning fossil fuels** (coal, oil, natural gas) in power stations, vehicles, homes and industry. • **Deforestation** — fewer trees to remove CO_2 by photosynthesis; burning or rotting of cleared trees releases CO_2. • Cement manufacture (thermal decomposition of limestone).
Methane	• **Animal farming** — cattle produce methane during digestion. • **Decomposition of rubbish in landfill sites** (anaerobic decay). • **Rice paddy fields** (waterlogged soil bacteria). • Leaks from natural gas extraction and pipelines.

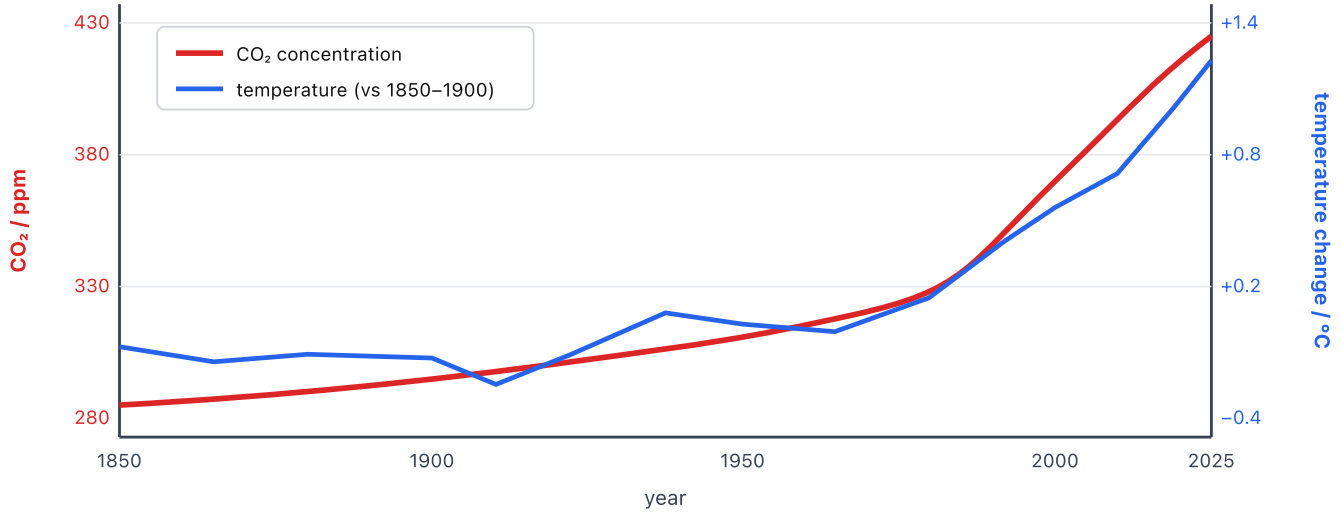
The Evidence and Its Uncertainties

Based on **peer-reviewed evidence**, many scientists believe that human activities will cause the temperature of the Earth's atmosphere to increase at the surface and that this will result in **global climate change**.

However, it is difficult to model such complex systems as global climate change. This leads to **simplified models**, **speculation** and **opinions** presented in the media that may be based on only parts of the evidence and which may be **biased**.

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Atmospheric CO_2 and global temperature since 1850 (approximate)



Illustrative shape based on published records; a correlation like this does not on its own prove causation — that requires a mechanism and peer-reviewed models.

Evaluating the Evidence

- **Correlation:** CO_2 concentration and global temperature have both risen since the Industrial Revolution (about 1750–1850).
- **Mechanism:** there is a well-understood physical mechanism (absorption of infrared radiation by greenhouse gases) that explains *why* more CO_2 should cause warming.
- **Peer review:** scientific findings are checked by other scientists before publication, which increases confidence in the conclusions. The overwhelming majority of climate scientists agree that recent warming is mainly caused by human activity.
- **Uncertainty:** temperature varies naturally from year to year; older data (e.g. from ice cores) have greater uncertainty; climate models are simplifications with many variables; future emissions are unknown.
- **Media:** newspapers and websites may present simplified, selective or biased accounts.

Exam Tip: When asked to **evaluate** a claim about climate change, comment on: the **quality** of the data (sample size, time span, source), whether it shows **correlation or causation**, whether it has been **peer reviewed**, and whether the source may be **biased**. Use specific numbers from any graph you are given — "CO_2 rose from about 315 ppm to about 420 ppm" is much stronger than "it went up".

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4.9.2.3 Global Climate Change

An increase in average global temperature is a major cause of climate change. There are several potential effects of global climate change:

Effect	Details
Rising sea levels	Melting of polar ice caps and glaciers, and thermal expansion of seawater, may cause **flooding** of low-lying land and **increased coastal erosion**.
More frequent and severe storms	Extreme weather events (hurricanes, floods, heatwaves) become more common.
Changes in the amount, timing and distribution of rainfall	Some regions experience droughts, others flooding.
Temperature and water stress	For humans and wildlife — shortages of fresh water.
Changes in food-producing capacity	Crops may fail in some regions; some areas may no longer be able to grow certain crops.
Changes in distribution of wildlife species	Species migrate, are lost from some areas, or become extinct; pests and diseases may spread to new regions.

Students should be able to discuss the **scale, risk and environmental implications** of global climate change — for example, a sea-level rise of 1 m would affect hundreds of millions of people living in low-lying coastal cities and river deltas.

4.9.2.4 The Carbon Footprint and Its Reduction

> The **carbon footprint** is the **total amount of carbon dioxide and other greenhouse gases emitted over the full life cycle** of a product, service or event.

The carbon footprint can be reduced by reducing emissions of carbon dioxide and methane:

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Method	How it reduces the carbon footprint
Increased use of alternative energy supplies	Wind, solar, nuclear, hydroelectric — generate electricity without burning fossil fuels.
Energy conservation	Insulating homes, using efficient appliances and LED lighting, using public transport — less fuel burned overall.
Carbon capture and storage (CCS)	CO_2 from power stations is captured and pumped deep underground into old oil/gas fields or porous rocks instead of being released.
Carbon taxes and licences	Make it more expensive to emit CO_2, encouraging companies and individuals to emit less.
Carbon off-setting, including through tree planting	Paying for projects (e.g. planting forests) that remove an equivalent amount of CO_2 from the atmosphere by photosynthesis.
Carbon-neutral fuels and processes	Using fuels where no *net* CO_2 is released (e.g. biofuels, where the CO_2 released was recently absorbed by the plants).

Problems in Reducing the Carbon Footprint

Actions may be **limited** for several reasons:

- **Scientific disagreement** about the causes and consequences of global climate change (though this is now small).
- **Lack of public information and education**.
- **Lifestyle changes** are unpopular — e.g. flying less, eating less meat, using cars less.
- **Economic considerations** — alternatives can be expensive; developing countries may need cheap fossil fuels to grow their economies.
- **Incomplete international cooperation** — one country cutting emissions has little effect if others do not.

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Worked Example 2 — Comparing Carbon Footprints

Question: Burning 1 kg of methane releases 2.75 kg of CO_2. Burning 1 kg of coal (assume pure carbon) releases how much CO_2? Which fuel has the bigger carbon footprint per kg? (Ar: C = 12, O = 16)

Coal: $C + O_2 \rightarrow CO_2$
12 g of C → 44 g of CO_2
1 kg of C → 44 ÷ 12 = 3.67 kg of CO_2

Methane: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
16 g of CH_4 → 44 g of CO_2
1 kg of CH_4 → 44 ÷ 16 = 2.75 kg of CO_2 ✓

Answer: Coal releases 3.67 kg CO_2 per kg, compared with 2.75 kg for methane, so coal has the bigger carbon footprint per kg burned. (Methane also releases more energy per kg, making the difference per kJ even larger.)

Exam Tip: "Carbon neutral" does not mean "produces no CO_2". It means the CO_2 released is **balanced** by CO_2 absorbed. In practice biofuels are rarely fully carbon neutral, because energy (often from fossil fuels) is used to grow, harvest, process and transport them — mention this for an evaluation mark.

4.9.3 Common Atmospheric Pollutants and Their Sources

4.9.3.1 Atmospheric Pollutants from Fuels

The **combustion of fuels** is a major source of atmospheric pollutants. Most fuels, including coal, contain **carbon and/or hydrogen** and may also contain some **sulfur**.

The gases released into the atmosphere when a fuel is burned may include **carbon dioxide**, **water vapour**, **carbon monoxide**, **sulfur dioxide** and **oxides of nitrogen**. Solid particles and unburned hydrocarbons may also be released that form **particulates** in the atmosphere.

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Pollutant	How it is formed	Example equation
Carbon monoxide (CO)	**Incomplete combustion** of fuels containing carbon, when there is a limited supply of oxygen.	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$
Soot / carbon particulates (C)	Incomplete combustion with very limited oxygen; unburned hydrocarbons and solid particles (especially from diesel engines).	$CH_4 + O_2 \rightarrow C + 2H_2O$
Sulfur dioxide (SO_2)	**Sulfur impurities** in the fuel (especially coal) react with oxygen when the fuel burns.	$S + O_2 \rightarrow SO_2$
Oxides of nitrogen (NO_x: NO, NO_2)	**Nitrogen and oxygen from the air** react at the **very high temperatures** inside car engines and furnaces. (The nitrogen does *not* come from the fuel.)	$N_2 + O_2 \rightarrow 2NO$ $2NO + O_2 \rightarrow 2NO_2$

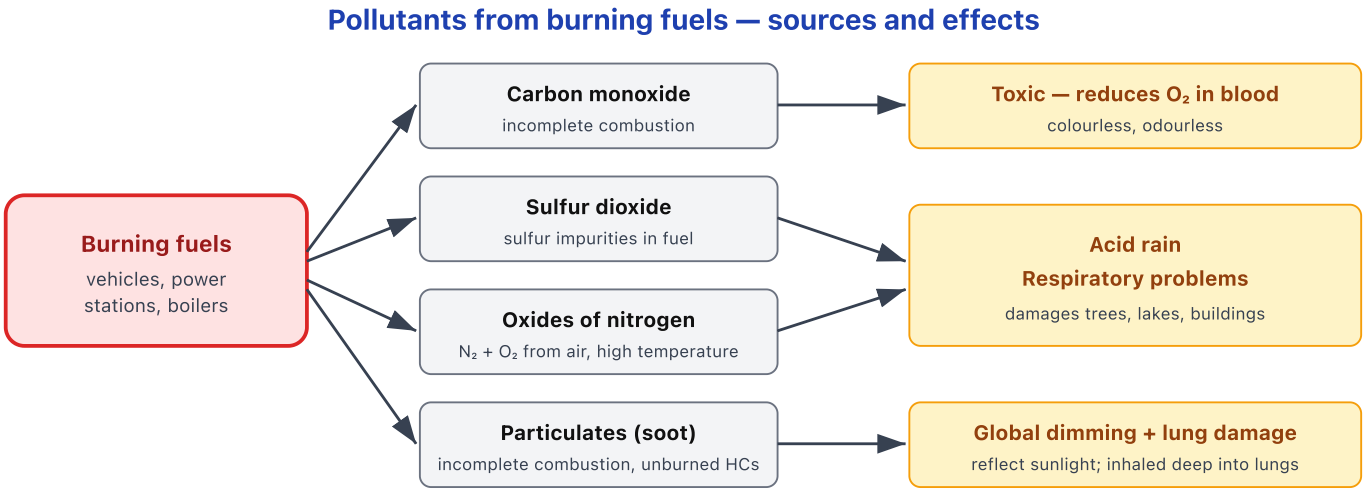
Exam Tip: A very common error: saying oxides of nitrogen come from "nitrogen in the fuel". They come from **nitrogen in the air** reacting with **oxygen in the air** at the **high temperature** in the engine. Similarly, sulfur dioxide comes from sulfur **in the fuel**, not in the air.

4.9.3.2 Properties and Effects of Atmospheric Pollutants

Pollutant	Effects
Carbon monoxide	A **toxic** gas. It is **colourless and odourless** and so is **not easily detected**. It combines with haemoglobin in red blood cells, reducing the blood's ability to carry oxygen — causing drowsiness, unconsciousness and death. Faulty gas boilers are a common source indoors; CO detectors are used in homes.

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Sulfur dioxide and **oxides of nitrogen**	Cause **respiratory problems** in humans (irritate the lungs; worsen asthma). They dissolve in water droplets in clouds to cause **acid rain**, which damages trees and plants, makes lakes and rivers too acidic for aquatic life, and corrodes limestone buildings, statues and metal structures.
Particulates	Cause **global dimming** — they reflect sunlight back into space, so less light reaches the Earth's surface. Small particles can be breathed deep into the lungs, causing **health problems** (lung and heart disease).



Reducing Pollutants

Although not a separate spec point, these methods are useful for evaluation answers:

- **Catalytic converters** in car exhausts convert CO to CO_2 and NO_x to N_2 (e.g. $2CO + 2NO \rightarrow 2CO_2 + N_2$).
- **Removing sulfur** from fuels before they are burned (low-sulfur petrol and diesel).
- **Flue-gas desulfurisation** ("scrubbers") at power stations, using calcium carbonate/calcium oxide to remove SO_2.
- **Particulate filters** on diesel vehicles.
- Ensuring a good **air supply** (e.g. servicing boilers) to promote complete combustion.

Worked Example 3 — Incomplete Combustion Equations

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Question: Write balanced equations for the combustion of propane, C_3H_8, producing (a) carbon monoxide and water, and (b) carbon (soot) and water.

(a) C: 3 C → 3CO H: 8 H → $4H_2O$
O on right = 3 + 4 = 7 atoms → 3.5 O_2 → double everything
$2C_3H_8 + 7O_2 \rightarrow 6CO + 8H_2O$

(b) C: 3 C → 3C H: 8 H → $4H_2O$
O on right = 4 atoms → 2 O_2
$C_3H_8 + 2O_2 \rightarrow 3C + 4H_2O$

Compare with complete combustion: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
Less oxygen available → more CO and C, less energy released.

Exam Tip: Incomplete combustion releases **less energy** than complete combustion for the same amount of fuel, as well as producing toxic CO and soot. A yellow, smoky flame indicates incomplete combustion; a blue flame indicates complete combustion. This links back to 4.7 (alkenes burn with smoky flames).

Summary Checklist for 4.9 Chemistry of the Atmosphere

- State the approximate composition of today's atmosphere (~80% N_2, ~20% O_2, small amounts of CO_2, water vapour and noble gases) and that it has been stable for ~200 million years.
- Describe one theory of the early atmosphere (volcanic activity, mainly CO_2, little or no O_2, like Mars and Venus) and explain why evidence is limited.
- Explain how the oceans formed and how CO_2 dissolved and formed carbonate sediments.
- Explain how oxygen increased (algae ~2.7 billion years ago, then plants) and write the photosynthesis equation.
- Explain how CO_2 decreased (oceans, photosynthesis, limestone, coal, crude oil and natural gas).
- Name the greenhouse gases (water vapour, CO_2, methane) and explain the greenhouse effect in terms of short- and long-wavelength radiation.

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- Describe human activities that increase CO_2 and methane levels.
- Evaluate evidence for climate change: peer review, correlation vs causation, uncertainty, simplified models and media bias.
- Describe the potential effects of global climate change and discuss scale, risk and environmental implications.
- Define carbon footprint; describe ways to reduce it and why action may be limited.
- Describe how CO, soot/particulates, SO_2 and NO_x are produced; write equations for complete and incomplete combustion.
- Describe and explain the effects of CO (toxic, undetectable), SO_2/NO_x (respiratory problems, acid rain) and particulates (global dimming, health).

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